

ANALYSIS OF BATTERY VOLTAGE UNDER LOAD

OVERVIEW

Batteries have inherent limitations due to their internal resistance, which causes voltage to drop as current increases. The maximum continuous current each battery can safely provide constrains how much load it can drive while maintaining functional voltage for connected devices.

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Analysis of Battery Voltage Under Load: Research Project

An applied study of battery performance in electrical systems, analyzing how internal resistance and current limits affect voltage delivery, system reliability, and safe operation under varying load conditions.

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Acronyms Used

American Society of Mechanical Engineers (ASME) 1
Eastern Washington University (EWU) 1
Nickel-Metal Hydride (NiMH) 1

Project Background

This project investigates the electrical behavior of battery packs in a small robotic system, specifically for a robotic device designed by Eastern Washington University's (EWU) American Society of Mechanical Engineers (ASME) chapter to compete in the national eFest competition. The goal was to determine a battery configuration that reliably supplies sufficient current while maintaining voltage above a critical minimum. The study examines three types of batteries: Alkaline 9V, Alkaline AA, and Nickel-Metal Hydride (NiMH) AA. It also analyzes how internal resistance, series and parallel arrangements, and maximum current limits influence system performance.

Project Goal

The goal was to design a battery pack capable of powering a brushless motor and a couple low-draw components that run on 5 V through a buck converter, while minimizing the total number of cells to conserve space and weight. The target max constant current load was 2 A, which represents the estimated maximum current required by the electronics in the robot. A minimum operating voltage of 6 V was required to ensure the buck converter could deliver a stable 5 V to the Arduino and servos. The design explores series and parallel configurations to understand how each arrangement affects voltage under load and maximum sustainable current.

Key Concepts & Functionality

The study considers batteries as real-world voltage sources with inherent limitations. Each cell has a nominal voltage and internal resistance that reduces terminal voltage under load. Packs in series increase total voltage while packs in parallel increase maximum current capacity. The effective output voltage V_{load} can be modeled by:

$$V_{\text{load}} = V_{\text{nominal}} - I_{\text{draw}} \cdot R_{\text{internal}}$$

where I_{draw} is the load current and R_{internal} is the combined internal resistance of the pack. Maximum continuous current for a battery or pack is capped by manufacturer ratings, ensuring safe operation without overheating or rapid voltage collapse.

Minimum operating voltage is determined by downstream electronics; for example, microcontrollers and peripherals often require regulated inputs through a buck converter. If the pack voltage drops below this threshold under load, devices may malfunction or reset. Graphically, this is represented by a region below the minimum voltage, illustrating the unsafe operating zone.

Series vs. Parallel

Optimizing a battery pack involves balancing voltage, current capacity, and the number of cells (space required to house). Series configurations raise voltage, enabling devices to meet input thresholds with fewer parallel cells, while parallel configurations allow the pack to supply higher current without exceeding individual cell ratings. For a given target current and voltage, the ideal configuration is the combination that satisfies both constraints with the least total cells, reducing cost, weight, and complexity.

Battery Characteristics and Rationale

Table 1 summarizes the fundamental electrical properties of each battery chemistry and the pack configurations theoretically tested. It provides a concise reference for comparing nominal voltage, internal resistance, and maximum current capabilities, illustrating why certain chemistries are better suited to the project's load requirements.

Table 1. Battery Chemistry & Fundamental Limits (Hibpshman, 2026; Learning about Electronics).

Feature Parameter	**Alkaline 9V**	**Alkaline AA**	**NiMH AA**
Nominal Voltage per Cell	9 V	1.5 V	1.2 V
Internal Resistance per Cell	1.5 Ω	0.15 Ω	0.02 Ω
Maximum Continuous Current	0.1 A	1 A	3 A
Tested Pack Configuration	1 cell	6 series (6 cells)	6 series (6 cells)
Effective Nominal Voltage	9 V	9.0 V	7.2 V
Effective Internal Resistance	1.5 Ω	0.9 Ω	0.12 Ω

Alkaline 9V

The Alkaline 9V cell has 1.5 Ω internal resistance and 0.1 A current limit. At a 2 A load, the terminal voltage would drop to:

$$V_{\text{load}} = 9 - 2 \cdot 1.5 = 6\ \text{V}$$

Even though it nominally hits the minimum threshold, the current far exceeds the 0.1 A maximum, making this battery unsafe and unsuitable for motor operation. The high internal resistance limits current output and causes significant voltage drop for modest loads.

Alkaline AA

Six AA alkaline cells in series provide 9 V nominal and 0.9 Ω total internal resistance. At a 2 A load:

$$V_{\text{load}} = 9 - 2 \cdot 0.9 = 7.2\ \text{V}$$

This comfortably exceeds the 6 V minimum. Series connection ensures adequate voltage, but each AA can supply only 1 A continuously. The motor's 2 A requirement slightly exceeds individual cell ratings. In practice, the pack may function briefly, but careful current limiting is recommended.

NiMH AA

Six NiMH AA cells in series yield 7.2 V nominal and 0.12 Ω total resistance. At 2 A:

$$V_{\text{load}} = 7.2 - 2 \cdot 0.12 = 6.96\ \text{V}$$

This remains safely above the 6 V minimum while each cell easily supports 3 A. The low internal resistance and high current capability make NiMH AA cells the most reliable choice for continuous motor operation.

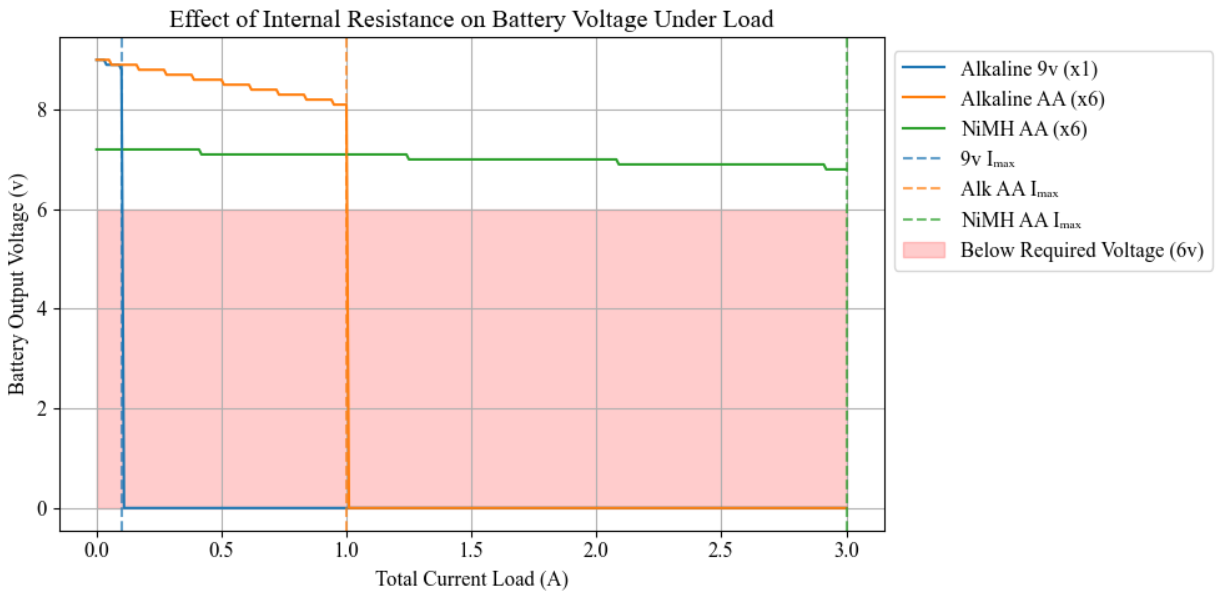
Visualizing Data

The graph of battery voltage versus current draw illustrates the effect of internal resistance and maximum current limits on each battery chemistry under load (Figure 1). The graph was created with the matplotlib Python library to iterate over many currents and how that would affect the output voltage given each batteries fundamental properties. Each curve slopes downward, reflecting Ohm's law:

$$V_{\text{load}} = V_{\text{nominal}} - I \cdot R_{\text{internal}}$$

As current increases, the terminal voltage drops proportionally to the total internal resistance of the pack.

Figure 1. Effect of Internal Resistance on Battery Voltage Under Load (Hibpshman, 2026).



Conclusion Supported by Figure 1

For the Alkaline 9V, the voltage decline demonstrates its internal resistance of 1.5 Ω. At the motor's 2 A load, the terminal voltage reaches 6 V, the absolute minimum required. Combined with the cell's 0.1 A current limit, this confirms the battery is unsafe and unsuitable for motor operation.

The Alkaline AA series pack exhibits a moderate voltage drop. At 2 A, terminal voltage is 7.2 V, safely above the minimum. While the voltage is adequate, each AA cell is rated for only 1 A continuous current, so the 2 A motor load slightly exceeds the per-cell limit. Practical use would require current limiting to avoid overstressing the cells.

The NiMH AA series pack shows a gentle decline due to very low internal resistance (0.12 Ω) and higher per-cell current rating (3 A). At 2 A, terminal voltage remains near 6.96 V, safely above the minimum. This chemistry is the most robust for continuous operation, supplying the motor reliably without approaching cell limits, though it may be slightly less space-efficient than Alkaline.

Shading below 6 V on the graph indicates unsafe operating voltage for the electronics, visually emphasizing which battery options are viable. Vertical lines at each battery's maximum rated current clarify operational boundaries, showing where each chemistry can safely deliver power. This graphical representation validates the numerical calculations, linking theory to practical performance under load.

References

Hibpshman, S. (2026, Feburary). Cheney, Washington, United States of America.

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